HAZE ALERT

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INTRODUCTION TO SDG13 AND PROBLEM STATEMENT

WHAT IS SDG 13 ?

Sustainable Goal 13 also known as SDG 13 is one of the 17 Sustainable Development Goals set by the UN to fight Climate Change.

In simple words, SDG 13 aims to urgent action against climate change and it’s effects to preserve the planet for future generations. Our main focus is Target 13.3 which is to improve education, awareness, and capacity on climate change mitigation, adaptation, impact reduction, and early warning systems.

CASE STUDY:
HAZE IN MALAYSIA

Malaysia in SE Asia regularly faces haze once a year between July and October. Farmers in neighboring Indonesia burn their agricultural waste and the wind direction carries the smog towards Western (Peninsular) Malaysia causing haze.

Haze can cause eye and throat infection. It is also harmful for people with respiratory issues such as asthma. Malaysian government Air readings, MyIPU app along with other measures to monitor the haze and inform people about it. Yet this has proved inefficient on several occasions leaving room for suggestions and improvement.

ABOUT HAZEALERT

HAZE
ALERT

HazeAlert is a computer program designed to increase public awareness of haze in Malaysia by disseminating crucial information about the Air Pollution Index (API) and to report haze-related incidents. It serves as a one-stop platform to tackle recurring haze in Malaysia

HAZEALERT FEATURES

Announcements

1. Public Health Advisory: Elevated Air Pollution Levels Detected
2. Air Quality Alert: Poor Air Conditions Forecasted
3. Community Action Needed: Combatting Air Pollution
4. Air Quality Improvement Efforts Underway
5. Environmental Awareness Workshop: Understanding Air Pollution

- Real-time dissemination of critical haze updates, warnings
- It serves as an information hub, ensuring timely alerts and measures are communicated to both government and public users

```
Q: What is air pollution?
A: Air pollution is the presence of substances in the air that

Q: How is air quality measured?
A: Air quality is typically measured using the Air Quality Inde

Q: What are the common causes of air pollution?
A: Common causes include emissions from vehicles, industrial pr

Q: How does weather affect air pollution?
A: Weather can significantly impact air pollution levels. For e

Q: What are the health effects of air pollution?
A: Exposure to polluted air can lead to respiratory diseases, h

Q: How can I protect myself from air pollution?
A: Staying informed about air quality, minimizing outdoor activ

Q: What is the greenhouse effect?
A: The greenhouse effect is a natural process where certain gas

Q: How can individuals help reduce air pollution?
A: Individuals can help by reducing vehicle use, conserving ene

Enter anything to exit:
```

FAQs

- A tab dedicated to demystifying haze by providing verified information.
- It addresses frequently asked questions, offering clarity on the impacts of haze.
- Users can access this repository to understand haze-related terms, impacts, and Do’s and Don’ts for informed decision-making

API Readings

- A crucial feature providing up-to-date air quality information based on the location of the particular user.
- This function includes methods to retrieve and display current API readings as per their location in the country
- This real-time data allows users to plan their outdoor activities accordingly

```
Day 1:
  Time 1: 75
  Time 2: 68
  Time 3: 72
  Time 4: 80
  Time 5: 92
  Time 6: 105
  Time 7: 112
  Time 8: 98
  Time 9: 85
  Time 10: 78
  Time 11: 65
  Time 12: 58
  Time 13: 60
  Time 14: 72
  Time 15: 88
  Time 16: 95
  Time 17: 103
  Time 18: 110
  Time 19: 98
  Time 20: 85
  Time 21: 76
  Time 22: 68
  Time 23: 72
  Time 24: 80
```

```
Day 2:
  Time 1: 78
  Time 2: 71
  Time 3: 73
  Time 4: 81
  Time 5: 93
  Time 6: 107
  Time 7: 115
  Time 8: 100
  Time 9: 87
  Time 10: 80
  Time 11: 66
  Time 12: 59
  Time 13: 62
  Time 14: 74
  Time 15: 89
  Time 16: 96
  Time 17: 104
  Time 18: 112
  Time 19: 99
  Time 20: 86
  Time 21: 78
  Time 22: 70
  Time 23: 74
  Time 24: 82
```

User Reports

```
1. It has come to our attention that the air quality in our co
2. I wanted to inform you that the air quality in Kuala Lumpur
3. It is with regret that we announce the closure of Taylor's
4. Considering the circumstances, I recommend limiting outdoor
5. Today, during my exploration of our city, I couldn't help b
```

- This function enables users to contribute real-time data on climate-related incidents, observations, and concerns.
- It enables community engagement by allowing individuals to share any observations or haze-related incidents.

User Profile

```
Please enter a valid username:
taylor00
Username accepted.
Please enter a valid password:
taylor
Password must be at least 8 characters long
00taylor
Password accepted.
Your account is almost complete. Next, please answer the following questions.
Please enter your email:
taylor
Invalid email address.
taylor@taylors.edu.my
Please enter your contact number:
abcdeft
Contact number should contain only digits.
0356789
Please enter your city:
Selangor
Your account has been successfully created. Welcome to the app taylor00
```

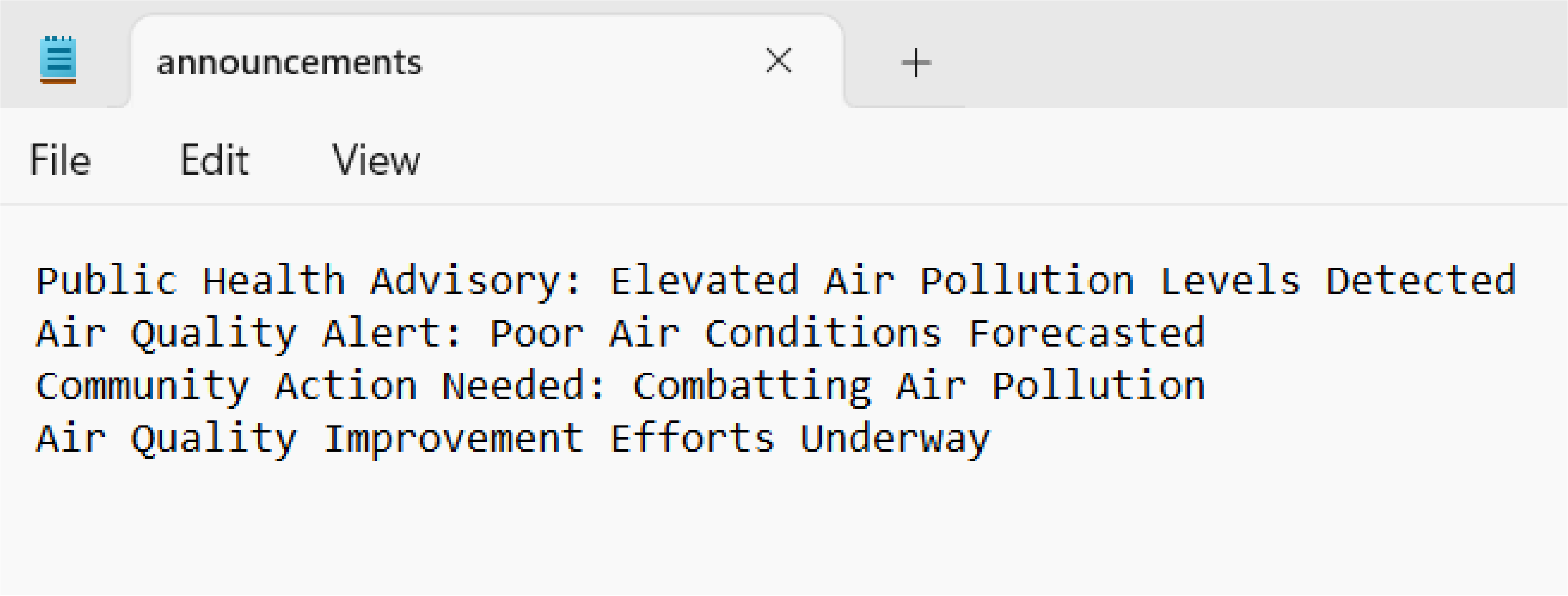
- This section allows users to create and manage their profiles.
- This feature manages user information offering the Government (Admin) an insight into user demographics.

TEXT FILE

```
private static String filePath = "announcements.txt";

public Announcements(){
    announcements = readDataFromFile(filePath);
}

private static String[] readDataFromFile(String filePath){
    try (BufferedReader reader = new BufferedReader(new
        FileReader(filePath))){
        return reader.lines().toArray(String[]::new);
    } catch (IOException e){
        e.printStackTrace();
        return new String[0];
    }
}
```



Public Health Advisory: Elevated Air Pollution Levels Detected
Air Quality Alert: Poor Air Conditions Forecasted
Community Action Needed: Combatting Air Pollution
Air Quality Improvement Efforts Underway

- Import and edit data from the text file

HAZEALERT CLASSES

ANNOUNCEMENTS

1. ADD ANNOUNCEMENT
2. EDIT ANNOUNCEMENET
3. REMOVE ANNOUNCEMENT

OUTPUT

```
1. Public Health Advisory: Elevated Air Pollution Levels Detected
2. Air Quality Alert: Poor Air Conditions Forecasted
3. Community Action Needed: Combatting Air Pollution
4. Air Quality Improvement Efforts Underway
5. Environmental Awareness Workshop: Understanding Air Pollution
```

```
1. View Announcements
2. Edit Announcement
3. Delete Announcement
4. Add Announcement
5. Exit
4
-----------------------------------
Please enter what announcement you would like create:
Taylors will not have school at Monday.
Successfully added announcement.
```

SOURCE CODE

```
public void addAnnouncement(String message) {
    // Shift existing announcements to make room for the new one
    for (int i = announcements.length - 1; i > 0; i--) {
        announcements[i] = announcements[i - 1];
    }
    announcements[0] = message;
    if (announcements.length == 11){
        announcements[10] = null;
    }
    System.out.println("Successfully added announcement.");
    writeDataToFile(filePath, announcements);
}
```

UTILISE A FOR LOOP TO SHIFT THE ANNOUNCEMENT 1 PLACE DOWN

ANNOUNCEMENTS

1. ADD ANNOUNCEMENT
2. EDIT ANNOUNCEMENET
3. REMOVE ANNOUNCEMENT

SOURCE CODE

```
public void editAnnouncement(int index, String newMessage) {
    if (index >= 0 && index < announcements.length) {
        announcements[index] = newMessage;
    } else {
        System.out.println("Invalid index for editing announcement.");
    }
    writeDataToFile(filePath, announcements);
}
```

OUTPUT

```
1. View Announcements
2. Edit Announcement
3. Delete Announcement
4. Add Announcement
5. Exit
2
-----------------------------------
-----------------------------------
1. Taylors will not have school at Monday.
2. Public Health Advisory: Elevated Air Pollution Levels Detected
3. Air Quality Alert: Poor Air Conditions Forecasted
4. Community Action Needed: Combatting Air Pollution
5. Air Quality Improvement Efforts Underway
Select the announcement number of the announcement you would like to edit:
1
You have selected announcement number 1. What would you like to change it to?
Taylors will not have school at next week.
Are you sure you want to change it to this? (y/n)
y
Successfully edited announcement. Returning to previous screen.
```

UTILISE A IF-ELSE STATEMENT TO CHECK THE ANNOUNCEMENT INDEX EXISTION

ANNOUNCEMENTS

1. ADD ANNOUNCEMENT
2. EDIT ANNOUNCEMENET
3. REMOVE ANNOUNCEMENT

SOURCE CODE

```
public void removeAnnouncement(int index) {
    if (index >= 0 && index < announcements.length) {
        // Shift announcements to remove the specified index
        for (int i = index; i < announcements.length - 1; i++) {
            announcements[i] = announcements[i + 1];
        }

        // Set the last element to null to remove it
        announcements[announcements.length - 1] = null;

    } else {
        System.out.println("Invalid index for removing announcement.");
    }

    System.out.println("Successfully deleted announcement.");
    writeDataToFile(filePath, announcements);
}
```

OUTPUT

```
1. Taylors will not have school at next week.
2. Public Health Advisory: Elevated Air Pollution Levels Detected
3. Air Quality Alert: Poor Air Conditions Forecasted
4. Community Action Needed: Combatting Air Pollution
5. Air Quality Improvement Efforts Underway
-----------------------------------
1. View Announcements
2. Edit Announcement
3. Delete Announcement
4. Add Announcement
5. Exit
3
-----------------------------------
-----------------------------------
1. Taylors will not have school at next week.
2. Public Health Advisory: Elevated Air Pollution Levels Detected
3. Air Quality Alert: Poor Air Conditions Forecasted
4. Community Action Needed: Combatting Air Pollution
5. Air Quality Improvement Efforts Underway
Which announcement would you like to remove? Enter 0 to exit.
1
Successfully deleted announcement.
```

FAQS

1. DISPLAY FAQS
2. ADD FAQS
3. DELETE FAQS

SOURCE CODE

```
public void displayFAQs() {
    for (int i = 0; i < questions.size(); i++) {
        System.out.println("Q: " + questions.get(i));
        System.out.println("A: " + answers.get(i));
        System.out.println();
    }
}
```

OUTPUT

```
1. View FAQ
2. Add FAQ
3. Delete FAQ
1
0) Q: What is air pollution?
0) A: Air pollution is the presence of substances in the air that are harmful to

1) Q: How is air quality measured?
1) A: Air quality is typically measured using the Air Quality Index (AQI), which

2) Q: What are the common causes of air pollution?
2) A: Common causes include emissions from vehicles, industrial processes, burnin

3) Q: How does weather affect air pollution?
3) A: Weather can significantly impact air pollution levels. For example, wind ca

4) Q: What are the health effects of air pollution?
4) A: Exposure to polluted air can lead to respiratory diseases, heart disease, s

5) Q: How can I protect myself from air pollution?
5) A: Staying informed about air quality, minimizing outdoor activities when poll

6) Q: What is the greenhouse effect?
6) A: The greenhouse effect is a natural process where certain gases in Earth's a

7) Q: How can individuals help reduce air pollution?
7) A: Individuals can help by reducing vehicle use, conserving energy, recycling,
```

FAQS

1. DISPLAY FAQS
2. ADD FAQS
3. DELETE FAQS

SOURCE CODE

```
public void addFAQ(String question, String answer){
    questions.add(question);
    answers.add(answer);
}
```

OUTPUT

```
1. View FAQ
2. Add FAQ
3. Delete FAQ
2
What question would you like to add:
What this program for?
What would the answer to that question be:
This program main purpose is letting the user know the air condition of the day, c
```

ITERATES THROUGH THE QUESTIONS LIST AND PRINTS EACH QUESTION FOLLOWED BY ITS CORRESPONDING ANSWER USING A FOR LOOP.

FAQS

1. DISPLAY FAQS
2. ADD FAQS
3. DELETE FAQS

SOURCE CODE

```
public void deleteFAQ() {
    Scanner scanner = new Scanner(System.in);

    //display the faqs
    displayFAQs();

    //user inputs which index to delete
    System.out.println("Enter the index of the FAQ you want to delete:");
    int indexToDelete = scanner.nextInt();

    //validity
    if (indexToDelete >= 0 && indexToDelete < questions.size()) {
        //remove faq (q and a) from index
        questions.remove(indexToDelete);
        answers.remove(indexToDelete);
        System.out.println("FAQ deleted successfully.");
    } else {
        System.out.println("Invalid index. Please enter a valid index.");
    }
}
```

OUTPUT

```
0) Q: What is air pollution?
0) A: Air pollution is the presence of substances in the air that are harmful to the health

1) Q: How is air quality measured?
1) A: Air quality is typically measured using the Air Quality Index (AQI), which quantifies

2) Q: What are the common causes of air pollution?
2) A: Common causes include emissions from vehicles, industrial processes, burning of fossi

3) Q: How does weather affect air pollution?
3) A: Weather can significantly impact air pollution levels. For example, wind can disperse

4) Q: What are the health effects of air pollution?
4) A: Exposure to polluted air can lead to respiratory diseases, heart disease, stroke, and

5) Q: How can I protect myself from air pollution?
5) A: Staying informed about air quality, minimizing outdoor activities when pollution leve

6) Q: What is the greenhouse effect?
6) A: The greenhouse effect is a natural process where certain gases in Earth's atmosphere

7) Q: How can individuals help reduce air pollution?
7) A: Individuals can help by reducing vehicle use, conserving energy, recycling, and suppo

8) Q: What this program for?
8) A: This program main purpose is letting the user know the air condition of the day, and

Enter the index of the FAQ you want to delete:
8
FAQ deleted successfully.
```

UTILISE AN IF-ELSE STATEMENT TO CHECK THE VALIDITY

REPORTS

1. ADD REPORTS
2. EDIT REPORTS
3. DELETE REPORTS

SOURCE CODE

```
public void add(){

    //insert report
    System.out.println("What do you want to report?\n"
        + "[If you want to cancel, please key in 'exit']");
    reportInput=scan.nextLine();

    if(reportInput.equals("exit")){
      //back to main
      return;
    }
    else if(reports[0] == null){
      //input to the first
      reports[0] = reportInput;
    } else{
      //push down array items, then overwrite
      for(reportIndex= reports.length - 1; reportIndex > 0 ;reportIndex--){
        reports[reportIndex]=reports[reportIndex - 1];
      }
      reports[0] = reportInput;
      System.out.println("Thank you for the report");
    }
    writeDataToFile(filePath, reports);
  }
```

OUTPUT

```
1. It has come to our attention that the air quality in our community has be
ution, especially those with respiratory conditions.
2. I wanted to inform you that the air quality in Kuala Lumpur has recently
 for us to discuss and explore community-led initiatives to address this iss
3. It is with regret that we announce the closure of Taylor's today due to t
4. Considering the circumstances, I recommend limiting outdoor activities an
orities. Let's remain vigilant and prioritize our health.
5. Today, during my exploration of our city, I couldn't help but notice a di
bers of this community, perhaps it's time to collectively explore sustainabl
What do you want to report?
[If you want to cancel, please key in 'exit']
Dear God, oh dear! It is risky to drive in Cheras due to the strong haze, wh
Thank you for the report
```

RESULT:

```
1. Dear God, oh dear! It is risky to drive in Cheras due to the strong haze,
2. It has come to our attention that the air quality in our community has be
ution, especially those with respiratory conditions.
3. I wanted to inform you that the air quality in Kuala Lumpur has recently
 for us to discuss and explore community-led initiatives to address this iss
4. It is with regret that we announce the closure of Taylor's today due to t
5. Considering the circumstances, I recommend limiting outdoor activities an
orities. Let's remain vigilant and prioritize our health.
```

- UTILISE A IF-ELSE STATEMENT TO FILTER THE ACTION THAT THE USER TAKING.
    - EXIT -- BACK TO MAIN PAGE
    - REPORT -- INSERT TO THE ARRAY AND THE TEXT FILE

REPORTS

1. ADD REPORTS
2. EDIT REPORTS
3. DELETE REPORTS

SOURCE CODE

```
public void editFromIndex(int index, String editedReport){
        reports[index] = editedReport; //replaces report at index to the
        editedReport
        writeDataToFile(filePath, reports); //rewrites the data in the text
        file to match with the information here
        }
}
```

OUTPUT

```
1. Dear God, oh dear! It is risky to drive in Cheras due to the strong haze, which
2. It has come to our attention that the air quality in our community has been nota
ution, especially those with respiratory conditions.
3. I wanted to inform you that the air quality in Kuala Lumpur has recently been a
 for us to discuss and explore community-led initiatives to address this issue.
4. It is with regret that we announce the closure of Taylor's today due to the dete
5. Considering the circumstances, I recommend limiting outdoor activities and stayi
orities. Let's remain vigilant and prioritize our health.
Select the report number of the report you would like to edit:
1
You have selected report number 1. What would you like to change it to?
The bad weather is causing a large number of people to become seriously ill with th
Are you sure you want to change it to this? (y/n)
y
Successfully edited report. Returning to previous screen.
```

- BY THE REPORTS[INDEX] = EDITEDREPORT;
    - MATCH IT TO THE INDEX AND EDIT THE TEXT INSIDE
    - INSERT THE TEXT THAT USER TYPE-IN

REPORTS

1. ADD REPORTS
2. EDIT REPORTS
3. DELETE REPORTS

SOURCE CODE

```java
public void removeIndex() {
        System.out.println("Which report do you want to remove?\n"
                + "[If you want to exit, please key in '0']");

        //scan if the input valid
        try{
                reportIndex=scan.nextInt();
                if(reportIndex==exitInt){
                        //back to main
                        return;

        }else if (reportIndex >= 0 && reportIndex <reports.length) {
                scan.nextLine(); // again, just to eat up the line.

                //Confirm deletion of selected index
                System.out.println("You have selected report number " +
                reportIndex + ". Do you wish to proceed with deletion?
                (y/n)");

                if (scan.nextLine().equals("y")){
                        //remove report from index
                        reports[reportIndex - 1] = null;
                        System.out.println("Successfully removed
                        report.");
                } else {
                        System.out.println("Cancelling");
                        return;
                }
        }
        removeNullItems();
    }catch (InputMismatchException e) {
        //invalid input
        System.out.println("Invalid index. Please enter a valid index.");
    }
        writeDataToFile(filePath, reports);
}
```

OUTPUT

```
1. It has come to our attention that the air quality in our communit
2. I wanted to inform you that the air quality in Kuala Lumpur has r
3. It is with regret that we announce the closure of Taylor's today
4. Considering the circumstances, I recommend limiting outdoor activ
5. Today, during my exploration of our city, I couldn't help but not
Which report do you want to remove?
[If you want to exit, please key in '0']
3
You have selected report number 3. Do you wish to proceed with delet
y
Successfully removed report.
```

RESULT:

```
1. It has come to our attention that the air quality in our communi
2. I wanted to inform you that the air quality in Kuala Lumpur has
3. Considering the circumstances, I recommend limiting outdoor acti
4. Today, during my exploration of our city, I couldn't help but no
5. ---
```

- UTILISE A TRY-CATCH STATEMENT TO CHECK VALIDITY
- UTILISE A IF-ELSE STATEMENT TO FILTER THE ACTION THAT THE USER TAKING.
  - EXITINT -- BACK TO MAIN PAGE
  - REPORT -- DELETE THE REPORT ACCORDINGLY
    - IF THE REPORT IS THE LAST ONE -- NULL
    - NOT THE LAST ONE -- PUSH ALL THE REPORT AFTER THE REMOVE 1 PLACE BEFORE.

AIR POLLUTION INDEX

1. ADD READING
2. DELETE READING
3. VIEWALLREADINGS
4. VIEWSPECIFICDAY

SOURCE CODE

```
// Add a new reading, and remove the oldest if necessary
public void addReading() {
        //shift down
        try {
            for (int i = readings.length; i > 0; i--) {
                readings[i] = readings[i - 1];
            }
        } catch (Exception ignored){

        }

        //add reading to the top
        for (int i = 0; i < readings[0].length; i++) {
            System.out.println("Enter the reading at time: " + i);
            readings[0][i] = scan.nextInt();

        }
        writeDataToFile(readings, filePath);
}
```

OUTPUT

```
Enter the reading at time: 0
76
Enter the reading at time: 1
75
Enter the reading at time: 2
88
Enter the reading at time: 3
63
Enter the reading at time: 4
68
Enter the reading at time: 5
71
Enter the reading at time: 6
5
Enter the reading at time: 7
69
Enter the reading at time: 8
77
Enter the reading at time: 9
81
Enter the reading at time: 10
74
Enter the reading at time: 11
76
```

```
Enter the reading at time: 12
75
Enter the reading at time: 13
79
Enter the reading at time: 14
69
Enter the reading at time: 15
77
Enter the reading at time: 16
75
Enter the reading at time: 17
74
Enter the reading at time: 18
73
Enter the reading at time: 19
72
Enter the reading at time: 20
71
Enter the reading at time: 21
77
Enter the reading at time: 22
73
Enter the reading at time: 23
66
```

UTILISED A FOR LOOP TO ENCRYPT THE RANGE OF READING LENGTH.

AIR POLLUTION INDEX

1. ADD READING
2. DELETE READING
3. VIEWALLREADINGS
4. VIEWSPECIFICDAY

SOURCE CODE

```
// Delete a reading
public void deleteReading(int day) {
        for(int i = day - 1; i < readings.length; i++){
            if (day == readings.length) {
                readings[day] = null;
        } else{
                readings[i] = readings[i + 1];
            }
        }
        System.out.println("Successfully deleted.");
        writeDataToFile(readings, filePath);
}
```

OUTPUT

```
1. View Readings
2. Edit Readings
3. Delete Readings
4. Add Readings
5. Exit
3
-----------------------------------
Day 1:
  Time 1: 1
  Time 2: 23
  Time 3: 39
  Time 4: 345
  Time 5: 32
  Time 6: 32
  Time 7: 34
  Time 8: 65786674
  Time 9: 33214323
  Time 10: 554322
  Time 11: 5434
  Time 12: 34553
  Time 13: 5553
  Time 14: 4432
  Time 15: 443
  Time 16: 554
  Time 17: 432
  Time 18: 5533
  Time 19: 443
  Time 20: 5531
  Time 21: 5432
  Time 22: 432
  Time 23: 567
  Time 24: 5423
Day 2:
```

```
Which day's readings would you like to delete
1
0. 1
1. 23
2. 39
3. 345
4. 32
5. 32
6. 34
7. 65786674
8. 33214323
9. 554322
10. 5434
11. 34553
12. 5553
13. 4432
14. 443
15. 554
16. 432
17. 5533
18. 443
19. 5531
20. 5432
21. 432
22. 567
23. 5423
Would you like to delete this day's readings? (y/n)
>>y
Successfully deleted.
```

UTILISED A FOR LOOP TO ENCRYPT THE RANGE OF READING LENGTH.

AIR POLLUTION INDEX

1. ADD READING
2. DELETE READING
3. VIEWALLREADINGS
4. VIEWSPECIFICDAY

SOURCE CODE

```java
// View all readings
public void viewAllReadings() {
    for (int day = 0; day < readings.length; day++) {
        System.out.println("Day " + (day + 1) + ":");
        for (int time = 0; time < readings[day].length; time++) {
            Integer reading = readings[day][time];
            String readingStr = reading != null ? reading.toString() :
            "No Reading";
            System.out.println("  Time " + (time + 1) + ": " +
            readingStr);
        }
    }
        writeDataToFile(readings, filePath);
}
```

OUTPUT

```
Day 1:
  Time 1: 75
  Time 2: 68
  Time 3: 72
  Time 4: 80
  Time 5: 92
  Time 6: 105
  Time 7: 112
  Time 8: 98
  Time 9: 85
  Time 10: 78
  Time 11: 65
  Time 12: 58
  Time 13: 60
  Time 14: 72
  Time 15: 88
  Time 16: 95
  Time 17: 103
  Time 18: 110
  Time 19: 98
  Time 20: 85
  Time 21: 76
  Time 22: 68
  Time 23: 72
  Time 24: 80
```

```
Day 2:
  Time 1: 78
  Time 2: 71
  Time 3: 73
  Time 4: 81
  Time 5: 93
  Time 6: 107
  Time 7: 115
  Time 8: 100
  Time 9: 87
  Time 10: 80
  Time 11: 66
  Time 12: 59
  Time 13: 62
  Time 14: 74
  Time 15: 89
  Time 16: 96
  Time 17: 104
  Time 18: 112
  Time 19: 99
  Time 20: 86
  Time 21: 78
  Time 22: 70
  Time 23: 74
  Time 24: 82
```

```
Day 3:
  Time 1: 80
  Time 2: 72
  Time 3: 74
  Time 4: 82
  Time 5: 94
  Time 6: 108
  Time 7: 116
  Time 8: 101
  Time 9: 88
  Time 10: 81
  Time 11: 67
  Time 12: 60
  Time 13: 63
  Time 14: 75
  Time 15: 90
  Time 16: 97
  Time 17: 105
  Time 18: 113
  Time 19: 100
  Time 20: 87
  Time 21: 79
  Time 22: 71
  Time 23: 75
  Time 24: 83
```

```
Day 4:
  Time 1: 120
  Time 2: 130
  Time 3: 140
  Time 4: 150
  Time 5: 160
  Time 6: 170
  Time 7: 180
  Time 8: 190
  Time 9: 200
  Time 10: 210
  Time 11: 220
  Time 12: 230
  Time 13: 240
  Time 14: 250
  Time 15: 260
  Time 16: 270
  Time 17: 280
  Time 18: 290
  Time 19: 300
  Time 20: 310
  Time 21: 320
  Time 22: 330
  Time 23: 340
  Time 24: 350
```

```
Day 5:
  Time 1: 360
  Time 2: 370
  Time 3: 380
  Time 4: 390
  Time 5: 400
  Time 6: 410
  Time 7: 420
  Time 8: 430
  Time 9: 440
  Time 10: 450
  Time 11: 460
  Time 12: 470
  Time 13: 480
  Time 14: 490
  Time 15: 500
  Time 16: 510
  Time 17: 520
  Time 18: 530
  Time 19: 540
  Time 20: 550
  Time 21: 560
  Time 22: 570
  Time 23: 580
  Time 24: 590
```

USE A FOR LOOP TO DISPLAYS ALL READINGS WITH THEIR RESPECTIVE DAYS AND TIMES.

AIR POLLUTION INDEX

1. ADD READING
2. DELETE READING
3. VIEWALLREADINGS
4. VIEWSPECIFICDAY

SOURCE CODE

```
public void viewSpecificDay(int day){
        day--; //day - 1 to make it a proper array index num
        int time = 0;
        for (int i : readings[day]){
            System.out.println(time + ". " + i);
            time++;
        }
        writeDataToFile(readings, filePath);
    }
}
```

OUTPUT

```
1. View Readings
2. Edit Readings
3. Delete Readings
4. Add Readings
5. Exit
1
Would you like to view the readings of a specific day? (y/n)
y
Please enter the day you would like to view (1 - 7)
2
0. 78
1. 71
2. 73
3. 81
4. 93
5. 107
6. 115
7. 100
8. 87
9. 80
10. 66
11. 59
12. 62
13. 74
14. 89
15. 96
16. 104
17. 112
18. 99
19. 86
20. 78
21. 70
22. 74
23. 82
```

USE A FOR LOOP TO DISPLAYS ALL READINGS WITH THEIR RESPECTIVE DAYS AND TIMES.

USER PROFILE

GET DATA FROM USER

SOURCE CODE

```
public String getUsername() {
        return username;
        }

        public void setUsername(String username) {
        // Validation for username (can't be null or empty)
        if (username != null && !username.isEmpty()) {
        this.username = username;
        System.out.println("Username accepted.");
        } else {
        throw new IllegalArgumentException("Username cannot be null or
        empty.");
        }
}

        public boolean checkPassword(String password){
        return (password.equals(this.password));
        }

        public void setPassword(String password){
        if(password.length() < 8){
        throw new IllegalArgumentException("Password must be at least 8
        characters long");
        } else {
        if (!password.isEmpty()) {
                this.password = password;
                System.out.println("Password accepted.");
        } else {
                throw new IllegalArgumentException("Password cannot be
                null");
        }
    }
}
```

OUTPUT

```
Please enter a valid username:
taylor00
Username accepted.
Please enter a valid password:
taylor
Password must be at least 8 characters long
00taylor
Password accepted.
Your account is almost complete. Next, please answer the following questions.
Please enter your email:
taylor
Invalid email address.
```

SOURCE CODE

OUTPUT

```
public String getPassword(){
return password;
}

public String getEmailAddress() {
return emailAddress;
}

public void setEmailAddress(String emailAddress) {
// Sanitize email by converting it to lowercase
emailAddress = emailAddress.toLowerCase();

// Validation for email using a simple regex pattern
if (isValidEmail(emailAddress)) {
this.emailAddress = emailAddress;
} else {
throw new IllegalArgumentException("Invalid email address.");
}
```

```
Please enter your email:
taylor
Invalid email address.
taylor@taylors.edu.my
```

SOURCE CODE

```
public String getContactNumber() {
return contactNumber;
}

public void setContactNumber(String contactNumber) {
// Validation for contact number (contains only digits)
if (isValidContactNumber(contactNumber)) {
this.contactNumber = contactNumber;
} else {
throw new IllegalArgumentException("Contact number should contain
only digits.");
}
}
```

OUTPUT

```
Please enter your contact number:
abcdeft
Contact number should contain only digits.
0356789
```

SOURCE CODE

OUTPUT

```
public String getCity() {
return city;
}

public void setCity(String city) {
// Validation for city (can't be null or empty)
if (city != null && !city.isEmpty()) {
this.city = city;
} else {
throw new IllegalArgumentException("City cannot be null or empty.");
}
}
```

```
Please enter your city:
Selangor
Your account has been successfully created. Welcome to the app taylor00
```

USE IF-ELSE STATEMENT TO CHECK VALIDITY

PROFILE

1. ADD PROFILE
2. DELETE PROFILE
3. UPDATE PROFILE

SOURCE CODE

```
public void addProfile(profile.Profile profile) {
if (profileExists(profile.getUsername())) {
throw new IllegalArgumentException("Username already
exists.");
}
profiles.add(profile);
writeDataToFile(filePath, profiles);
}

public boolean deleteProfile(String username) {
Iterator<profile.Profile> iterator = profiles.iterator();
while (iterator.hasNext()) {
profile.Profile profile = iterator.next();
if (profile.getUsername().equals(username)) {
        iterator.remove();
        return true; // Profile deleted successfully
}
}
writeDataToFile(filePath, profiles);
throw new IllegalArgumentException("Profile not found.");
}
```

OUTPUT

```
Welcome to the Air Readings Application. Please select an option to sign in.
Press 1 to log in
Press 2 to register an account
Press 3 to sign in as a guest
2
Please enter a valid username:
Joelle
Username already exists. Returning to login page.
0
Invalid Input, please try again.
Here are the options again: 1 - login, 2 - register account, 3 - sign in as guest
2
Please enter a valid username:
Joe
Username Accepted
Please enter a valid password:
12356789
Password accepted.
Your account is almost complete. Next, please answer the following questions.
Please enter your email:
joe@yahoo.com
Please enter your contact number:
0198654623
Please enter your city:
City cannot be null or empty. Try again.
Kuching
Your account has been successfully created. Welcome to the app Joe
```

```
3. Delete Profile
4. Exit
3
Please enter your current password:
12356789
Account deleted. Logging out.
```

PROFILE

1. ADD PROFILE
2. DELETE PROFILE
3. UPDATE PROFILE

SOURCE CODE

```
public boolean updateProfile(String username, profile.Profile
updatedProfile) {
for (int i = 0; i < profiles.size(); i++) {
profile.Profile profile = profiles.get(i);
if (profile.getUsername().equals(username)) {
        profiles.set(i, updatedProfile);
        return true; // Profile updated successfully
}
}
writeDataToFile(filePath, profiles);
throw new IllegalArgumentException("Profile not found.");
}

public static void registerUpdate(){
writeDataToFile(filePath, profiles);
}
```

OUTPUT

```
1. View Profile
2. Edit Profile
3. Delete Profile
4. Exit
2
Which element would you like to edit:
1. Username
2. Password
3. Email
4. Contact Number
5. City
6. Cancel
5
-----------------------------------
Please enter your current password:
12345678
Enter a new city:
Kuala Lumpur
Successfully changed city location.
```

OUTPUT HANDLERS

```java
public void showProfileOptions(Profile profile, ProfileManager
profileManager, boolean isGuest) {
        boolean toContinue = true;


        while (toContinue) {
        System.out.println("------------------------------------");

        if (isGuest) {
                // Display options for guests (login or register)
                System.out.println("1. Login");
                System.out.println("2. Register");
                System.out.println("3. Exit");

                try {
                int guestOption = scan.nextInt();

                switch (guestOption) {
                case 1:
                        //Guest login option
System.out.println("------------------------------------");
                        System.out.println("Guest Login");
                        profile =
registrationHandler.login(profileManager);
                        toContinue = false;
                        break;

                case 2:
                        //Guest register option

System.out.println("------------------------------------");
                        System.out.println("Guest Register");
                        profile =
registrationHandler.register(profileManager);
                        toContinue = false;
                        break;

                case 3:
                        toContinue = false;
                        break;

                default:
                        System.out.println("Invalid option. Please select
1, 2, or 3.");
                        break;
                }
                } catch (InputMismatchException e) {
System.out.println("Please input a valid number correlating to the
options given.");
                }
```

```
Welcome to the Air Readings Application. Please select an option to sign in.
Press 1 to log in
Press 2 to register an account
Press 3 to sign in as a guest
1
```

```
} else {
        // Display options for logged-in users
        System.out.println("1. View Profile");
        System.out.println("2. Edit Profile");
        System.out.println("3. Delete Profile");
        System.out.println("4. Exit");

        try {
        int optionKey = scan.nextInt();

        switch (optionKey) {
        case 1:
                user.viewProfile(profile);
                break;

        case 2:
                System.out.println("Which element would you
                like to edit:\n" +
                "1. Username\n" +
                "2. Password\n" +
                "3. Email\n" +
                "4. Contact Number\n" +
                "5. City\n" +
                "6. Cancel");
                int editKey = scan.nextInt();
                user.editProfile(profile, editKey);
                break;

        case 3:
                scan.nextLine();
                System.out.println("Please enter your current
                password:");
                String password = scan.nextLine();
                if (!profile.checkPassword(password)) {
                System.out.println("Incorrect password.");
                } else {
```

```
profileManager.deleteProfile(profile.getUsername());
    System.out.println("Account deleted. Logging out.");
    toContinue = false;
    }
    break;

            case 4:
                    toContinue = false;
                    break;

            default:
                    System.out.println("Invalid. Please correctly
                    enter which element you would like to edit.");
                    break;
            }
            } catch (InputMismatchException e) {
            System.out.println("Please input a valid number
            correlating to the options given.");
            }
        }
    }
}
```

```
1. View Profile
2. Edit Profile
3. Delete Profile
4. Exit
2
Which element would you like to edit:
1. Username
2. Password
3. Email
4. Contact Number
5. City
6. Cancel
```

MAIN METHOD

SOURCE CODE

```
public class App {
	static Readings readings;
	static Announcements announcements = new Announcements();
	static FAQ faq;
	static Reports reports = new Reports();
	static ProfileManager profileManager = new ProfileManager();
	static User user = new User();
	static OutputHandler outputHandler = new OutputHandler();
	static RegistrationHandler registrationHandler = new RegistrationHandler();

	public static void main(String[] args) throws Exception {
	Scanner scan = new Scanner(System.in);
	int tempInput;
	Profile currentUser = null;
	boolean isGuest = false;

	//Start Application: welcome message + sign in
	System.out.println(
		"Welcome to the Air Readings Application. Please select an option to
		sign in. \n"
		+ "Press 1 to log in \n"
		+ "Press 2 to register an account \n"
		+ "Press 3 to sign in as a guest"
	);
```

OUTPUT

```
Welcome to the Air Readings Application. Please select an option to sign in.
Press 1 to log in
Press 2 to register an account
Press 3 to sign in as a guest
```

UTILIZES THE REGISTRATIONHANDLER CLASS TO MANAGE USER SIGN-IN AND REGISTRATION PROCESSES.

SOURCE CODE

```
while (true){
        try {
                tempInput = scan.nextInt();
                break;
        } catch (InputMismatchException e) {
                System.out.println("Invalid input. Please enter an integer value.");
                scan.next();
        }
        }

        //Login or Registration options
        while (currentUser == null){
        switch (tempInput) {
                case 1:
                currentUser = registrationHandler.login(profileManager);
                break;
                case 2:
                currentUser = registrationHandler.register(profileManager);


                //Exit the loop
                break;
                case 3:

                //User guest sign in method
                System.out.println("Welcome Guest User.");
                currentUser = new Profile();
                isGuest = true;
                break;
```

OUTPUT

```
Input username:
user
Please enter the password:
01234567
Welcome User user
```

INPUTMISMATCHEXCEPTION TO MANAGE INCORRECT USER INPUTS.

SEARCH FEATURES

SOURCE CODE

```java
public void keywordSearch(Announcements announcements, Reports reports) {
        // User input for the keyword
        System.out.println("Enter keyword to search:");
        String keyword = scan.next();

        // Perform keyword search in announcements
        List<String> announcementResults = searchInList(announcements.getAnnouncements(), keyword, "Announcements");

        // Perform keyword search in reports
        List<String> reportResults = searchInList(reports.getReports(), keyword, "Reports");

        // Display search results
        System.out.println("Results from Announcements:");
        displaySearchResults(announcementResults);
        System.out.println("-----------------------------------");
        System.out.println("Results from User Reports");
        displaySearchResults(reportResults);
    }
```

OUTPUT

```
What would you like to view?
View Profile Options (1)
View Readings Page (2)
View Announcements Page (3)
View User Reports Page (4)
View FAQ (5)
Search (6)
Log Out and Close App (7)
6
Enter keyword to search:
Air
Results from Announcements:
Announcements: Public Health Advisory: Elevated Air Pollution Levels Detected
Announcements: Air Quality Alert: Poor Air Conditions Forecasted
Announcements: Community Action Needed: Combatting Air Pollution
Announcements: Air Quality Improvement Efforts Underway
-----------------------------------
Results from User Reports
No matching results found.
-----------------------------------
-----------------------------------
```

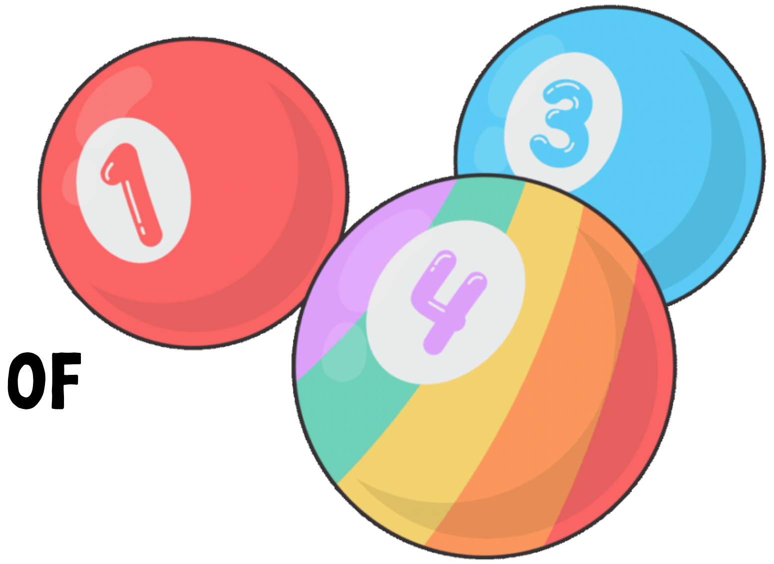
- UTILISE ARRAYLIST TO LIST OUT RESULT AND ANNOUNCEMENT
- UTILISE A IF-ELSE STATEMENT TO CHECK MATCHING RESULT

HAZEALERT USER FRIENDLY

How user friendly?

- User interface focuses on anticipating what users might need to do and ensuring that the interface has elements that are easy to access, understand, and use.
- This program is created in a way to create very clear and consice output.
- Outputs are handled by
  - OutputHandler class
    - show option of interaction between classes
  - App class
    - primary location of Main Method
    - ties classes together
- Try-catch statements used to catch errors

HOW USER FRIENDLY?



- CLEARLY DISPLAY WHAT USER CAN DO AND USE NUMBER AS A MEDIA OF LINKING USER INPUT AND PROGRAM

```
Welcome to the Air Readings Application. Please select an option to sign in.
Press 1 to log in
Press 2 to register an account
Press 3 to sign in as a guest
3
Welcome Guest User.
```


```
What would you like to view?
View Profile Options (1)
View Readings Page (2)
View Announcements Page (3)
View User Reports Page (4)
View FAQ (5)
Search (6)
Log Out and Close App (7)
```

HAZEALERT ERROR HANDLING

TRY...CATCH STATEMENTS

HazeAlert uses Command Line Interface where the input is taken from the user. However to prevent the program from crashing if wrong input is entered, HazeAlert uses **Try...Catch statements** and **Validating Inputs.**

```
---------------------------------------------------
Welcome to the Air Readings Application. Please select an option to sign in.
Press 1 to log in
Press 2 to register an account
Press 3 to sign in as a guest
testing
Invalid input. Please enter an integer value.
```

HOW DOES IT WORK ?

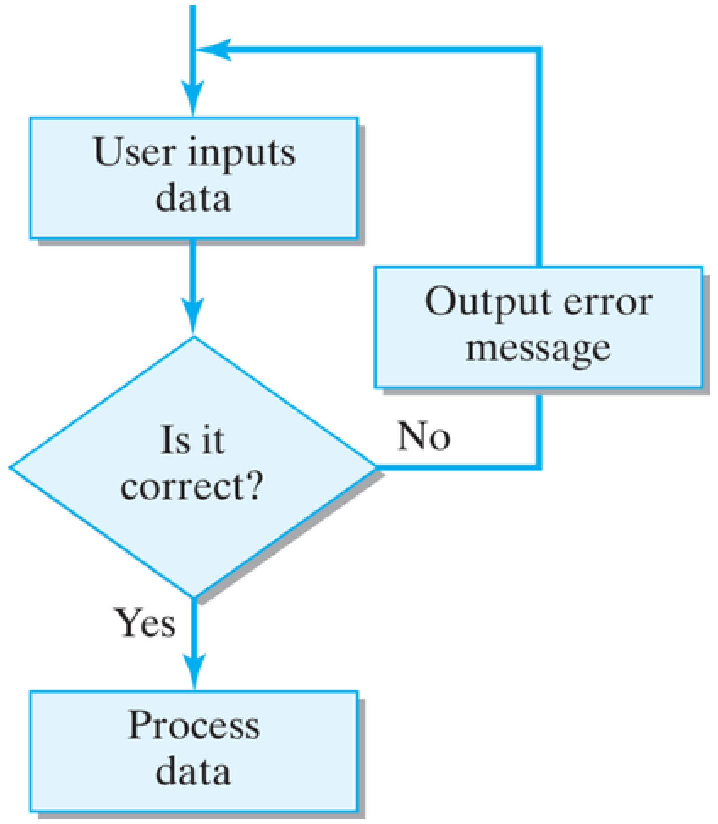
Notice that the user is only allowed a number of options (3 in this case). If the user inputs anything else the system will display “Invalid input. Please enter an integer value” without the program crashing.

VALIDATING INPUTS

It is important that the input entered by the user is checked with the requirements (validated).Examples are:

1. **Type Check -** check if input data is of required variable
2. **Presence Check -** check if user has placed an input or not and prevent null input.

Both of these are part of Try...Catch Statement.



RANGE CHECK & LENGTH CHECK ARE 2 OTHER EXAMPLES THAT ARE NOT PART OF TRY...CATCH STATEMENTS

RANGE CHECK

Range Check checks whether the user input falls within the valid range or not. If it doesn’t an error message is displayed and the program does not crash.

HOW DOES IT WORK ?

The user mut place an input within the valid range (in this case 1-7). If the user inputs something that is not within this range, it will display an error message “That is not a valid input” and the program does not crash.

```
What would you like to view?
View Profile Options (1)
View Readings Page (2)
View Announcements Page (3)
View User Reports Page (4)
View FAQ (5)
Search (6)
Log Out and Close App (7)
9
That is not a valid input.
-----------------------------------
What would you like to view?
View Profile Options (1)
View Readings Page (2)
View Announcements Page (3)
View User Reports Page (4)
View FAQ (5)
Search (6)
Log Out and Close App (7)
```

LENGTH CHECK

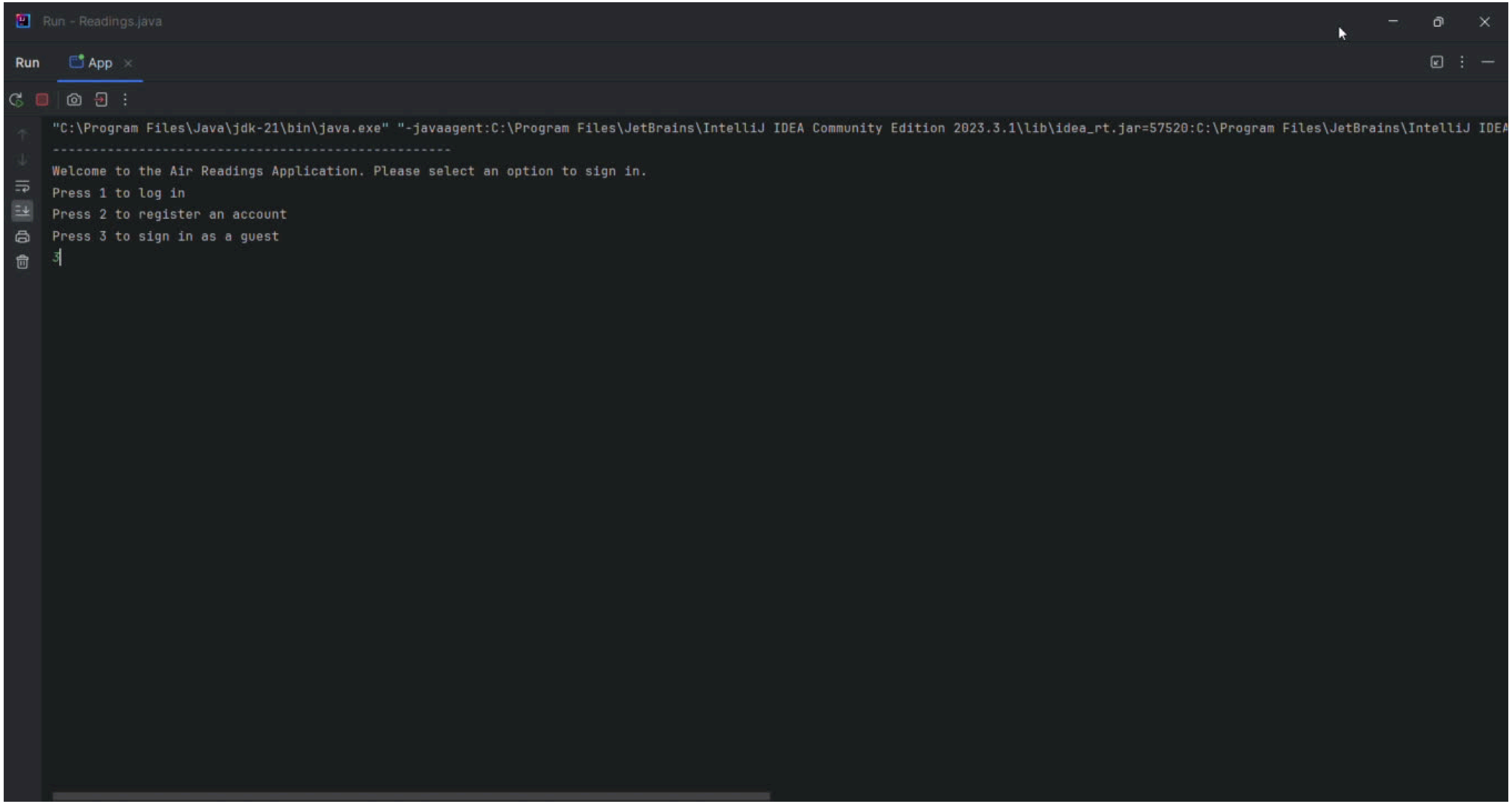
Length Check checks if the number of characters in the user's input meets the requirements set by the system. If it doesn't an error message is displayed and the program does not crash.

HOW DOES IT WORK ?

The user must input a password that is at least 8 characters long. If the input is less than that it will continue to display the error message until the user inputs a password that is at least 8 characters long.

```
-----------------------------------
Guest Register
Please enter a valid username:
usernameTest
Username Accepted
Please enter a valid password:
123
Password must be at least 8 characters long Try again.
1414
Password must be at least 8 characters long Try again.
1233456
Password must be at least 8 characters long Try again.
12345678
Password accepted.
```

PROGRAM DEMO


Run - Readings.java
Run
App
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2023.3.1\lib\idea_rt.jar=57520:C:\Program Files\JetBrains\IntelliJ IDEA

Welcome to the Air Readings Application. Please select an option to sign in.
Press 1 to log in
Press 2 to register an account
Press 3 to sign in as a guest
3

CONCLUSION

CONCLUSION

CONTRIBUTION TO SDG -13

1. Focus on Air Quality and Haze
2. Reflecting the Larger Climate Change Agenda
3. Features Supporting SDG 13

FUTURE IMPROVEMENTS

1. Data Visualization Enhancement
2. FAQs Section Improvement
3. Mobile Application Development